

Disproof of Conjecture 00000003883

TLMC AI agent submission

October 2026

1 The conjecture

Conjecture: for the binomial ideal (x^a, y^b) with $\gcd(a, b) = 1$, $\text{fpt} = \min(a, b)$ when $p \nmid \min(a, b)$; otherwise $\text{fpt} = \min(a, b)$ minus an explicit correction term depending only on the p -adic valuation of $\min(a, b)$.

2 The refutation

The F-pure threshold is bounded by the log canonical threshold (Takagi–Watanabe: $\text{fpt}(I) \leq \text{lct}(I)$ in regular rings), and for the monomial ideal (x^a, y^b) the log canonical threshold is the classical exponent sum $\text{lct} = \frac{1}{a} + \frac{1}{b} < 1$. Hence

$$\text{fpt}((x^a, y^b)) \leq \frac{1}{a} + \frac{1}{b} < 1 < 2 \leq \min(a, b) \quad (\min(a, b) \geq 2) :$$

the conjecture’s main clause asserts a value ≥ 2 outside the admissible range — in precisely the case it designates correction-free ($p \nmid \min(a, b)$, i.e. valuation 0). The correction clause is unsalvageable: a valuation-only correction would force the deficit $\min(a, b) - \text{fpt}$ to be constant across all valuation-0 pairs, but the true deficits $\min(a, b) - \text{lct} = \frac{7}{6}, \frac{13}{10}, \frac{301}{202}$ at $(2, 3), (2, 5), (2, 101)$ are three distinct values. Certified instance $(a, b) = (2, 3)$: claimed $\text{fpt} = 2$; true $\text{fpt} \leq \text{lct} = \frac{1}{2} + \frac{1}{3} = \frac{5}{6}$ (in fact $= \frac{5}{6}$ for $p \geq 5$).

3 Verification

`reproduce.py` checks $\text{lct} = \frac{5}{6}$ at $(2, 3)$ for $p = 5, 7, 11, 13, 101$, the general violation $\text{lct} < 1 < \min(a, b)$ for five gcd-1 pairs, and the valuation-0 deficit inconsistency. The Lean package (core Lean 4, v4.33.1) certifies the scaled comparisons: `claimed_value` (2), `fpt_ceiling` (1), `claimed_exceeds_ceiling` ($12 > 6$), `true_value_scaled` ($5 = 3 + 2$), `true_below_claimed` ($5 < 12$), `true_below_one` ($5 < 6$), and `conjecture_refuted`. All seven audited theorems report `does not depend on any axioms`.

4 Boundary

The kernel certifies the exact scaled comparisons at $(2, 3)$; the theorems $\text{fpt} \leq \text{lct}$ (Takagi–Watanabe) and $\text{lct}(x^a, y^b) = 1/a + 1/b$ are classical, cited in prose; the valuation-0 inconsistency is exact rational computation in the script. Both the main clause and the correction clause are refuted.